

THEMBISILE MAKHUBU

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EDUCATION

BSC COMPUTER SCIENCE

Jan '22 - Present

Stellenbosch University | Stellenbosch

NATIONAL SENIOR CERTIFICATE

Jan '17 - Dec '21

Falesizwe Secondary School | Frankfort

EXPERIENCE

Software Engineering Project Demonstrator

Jul '26 - Present

Department of Computer Science | Stellenbosch University

- Assess software engineering projects and student submissions against academic and quality standards, providing technical feedback on design, implementation, and engineering practices
- Support tutorials by assisting students with debugging, software design, and clean coding, while upholding the technical standards of the CS 344 Software Engineering module

Software Engineering Tutorial Demonstrator

Jul '25 - Nov '25

Department of Computer Science | Stellenbosch University

- Guided students through software engineering tutorials and practical exercises, assisting with debugging and understanding core software engineering concepts
- Supported collaborative problem-solving through practical activities while contributing to an engaging and supportive learning environment

Software Engineering Job Simulation on Forage

Sep '24

Standard Bank

- Engineered a Spring Boot application implementing JWT-based authentication and user access control
- Built an AWS Lambda function for image-based facial encoding as part of a mobile authentication workflow
- Extracted and analysed data from Teradata using SQL queries
- Developed a Django application providing personalised views of user data

Back-End Engineering Job Simulation on Forage

May '23 - May '23

Lyft

- Applied TDD, unit testing, modularisation, and software architecture principles to improve code reliability and maintainability
- Enhanced RESTful APIs for performance and scalability while using Git for version control and collaborative development

LEADERSHIP EXPERIENCE

Co-Founder & Chairperson

Jun '26 - Present

Stellenbosch Women in Information Technology (SWiIT) | Stellenbosch University

- Co-founded and lead SWiIT, a beginner-friendly society supporting women in IT, overseeing operations, events, partnerships, and communications
- Drive strategic initiatives and professional development while managing stakeholder relationships and SWiIT's digital presence across GitHub, LinkedIn, and Instagram

Class Representative

Mar '24 - Jun '24

Logistics Department | Stellenbosch University

- Represented student academic interests to university bodies and lecturers, gathering and communicating feedback to improve student-staff communication

Promoted collaboration through Microsoft Teams and supported effective communication between students and academic stakeholders	
PROJECTS	
TraceGuard	Jul '26 - Present
<i>Software Security & Risk Intelligence Platform</i>	
Developing a security and risk intelligence platform that analyses software repositories to detect vulnerabilities, exposed secrets, insecure configurations, and dependency risks Designing an interactive risk analysis platform that contextualises security findings across application architecture, prioritises remediation, and tracks security posture over time	
AlgoViz	Jun '26 - Jul '26
<i>- Interactive Algorithm Visualisation Platform</i>	
- Developed an interactive educational platform providing step-by-step visualisations to make algorithms easier to understand - Built using Next.js, TypeScript, Node.js, pnpm, and Tailwind CSS, with interactive real-time demonstrations - Applied component-based architecture, responsive web development, Git version control, and maintainable code organisation	
TECHNICAL SKILLS	
Languages: Java, TypeScript, JavaScript, Python, SQL, C, Assembly	
Frameworks & Technologies: Next.js, React, Node.js, Spring Boot, Django, Tailwind CSS, AWS Lambda	
Engineering: RESTful APIs, Git, JSON Web Tokens, Unit Testing, Test-Driven Development, Software Architecture	
Professional: Leadership, Communication, Problem-Solving, Adaptability, Time Management, Attention to Detail	
ACHIEVEMENTS	
Woman Regional Master (WRM)	Jun '25 - Present
<i>Chess SA Western Cape</i>	
Recognised by Chess South Africa for competitive excellence at regional level, demonstrating strategic thinking, focus, and sustained performance	
3rd Place – Standard Bank Group University Hackathon	Sep '25 -
<i>Standard Bank Cape Town</i>	
- Collaborated to develop, deploy, and present an AI-powered wellness solution within 48 hours - Developed multilingual communication and peer-connection functionality and integrated an AI coach providing personalised recommendations incorporating Standard Bank's value-added services	
Dell Young Leaders Scholarship Award	Feb '22 - Present
<i>Dell Young Leaders Stellenbosch</i>	
Selected for the scholarship in recognition of resilience, ambition, and leadership potential	
Top Performance – Computer Networking Projects	Feb '24 - Jun '24
<i>Stellenbosch University</i>	
Achieved an overall average of 88.44% across four networking projects: Client-Server Chat App (96.25%), P2P File Sharing App (93.75%), RBUDP vs TCP File Transfer (84.71%), and VoIP System (78.67%)	